

# EN.553.632 Bayesian Statistics

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# Chapter 1

## Introduction and Probability Review

### 1.1 Bayesian vs Frequentist Perspectives

**Definition 1.1.1.** Probability can be interpreted in two fundamentally different ways:

- **Frequentist:** Probability is the long-run relative frequency of an event.
- **Bayesian:** Probability is a degree of belief about an uncertain quantity.

In the Bayesian framework, unknown parameters are treated as random variables. This contrasts with the frequentist view, where parameters are fixed but unknown.

**Example 1.1.1.** Consider a coin flipped  $n = 10$  times with 7 heads observed.

- Frequentist estimate:  $\hat{\theta} = 0.7$
- Bayesian approach: place a prior  $\theta \sim \text{Beta}(\alpha, \beta)$

After observing  $y = 7$  heads, the posterior becomes:

$$\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

### 1.2 Bayes' Theorem

**Theorem 1.2.1** (Bayes' Rule). *For parameter  $\theta$  and data  $y$ ,*

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$

where

$$p(y) = \int p(y \mid \theta)p(\theta) \, d\theta$$

**Remark 1.2.1.** The posterior is often written up to proportionality:

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

## 1.3 Bernoulli and Binomial Models

### 1.3.1 Bernoulli Model

**Definition 1.3.1.** A Bernoulli random variable satisfies:

$$Y \sim \text{Bernoulli}(\theta), \quad P(Y = 1) = \theta$$

### 1.3.2 Binomial Model

**Definition 1.3.2.** For  $n$  independent Bernoulli trials,

$$Y \sim \text{Binomial}(n, \theta), \quad P(Y = y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

### 1.3.3 Likelihood

Given observations  $y_1, \dots, y_n$ , the likelihood is:

$$L(\theta) = \prod_{i=1}^n \theta^{y_i} (1 - \theta)^{1-y_i} = \theta^{\sum y_i} (1 - \theta)^{n - \sum y_i}$$

### 1.3.4 Maximum Likelihood Estimation

**Proposition 1.3.1.** The MLE for  $\theta$  in the Bernoulli/Binomial model is:

$$\hat{\theta}_{MLE} = \bar{y}$$

*Proof.* The log-likelihood is:

$$\ell(\theta) = \sum y_i \log \theta + (n - \sum y_i) \log(1 - \theta)$$

Differentiate:

$$\frac{d\ell}{d\theta} = \frac{\sum y_i}{\theta} - \frac{n - \sum y_i}{1 - \theta}$$

Set equal to zero:

$$\frac{\sum y_i}{\theta} = \frac{n - \sum y_i}{1 - \theta} \Rightarrow \theta = \frac{1}{n} \sum y_i$$

□

## 1.4 Conjugate Priors: Beta-Binomial Model

**Definition 1.4.1.** A prior is conjugate if the posterior belongs to the same family as the prior.

**Proposition 1.4.1.** *If*

$$\theta \sim \text{Beta}(\alpha, \beta), \quad Y \mid \theta \sim \text{Binomial}(n, \theta)$$

*then*

$$\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

*Proof.* Posterior:

$$p(\theta \mid y) \propto \theta^y (1 - \theta)^{n-y} \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Combine exponents:

$$p(\theta \mid y) \propto \theta^{\alpha+y-1} (1 - \theta)^{\beta+n-y-1}$$

which is the kernel of a Beta distribution. □

### 1.4.1 Posterior Mean

$$\mathbb{E}[\theta \mid y] = \frac{\alpha + y}{\alpha + \beta + n}$$

**Remark 1.4.1.** This can be interpreted as a weighted average:

$$\mathbb{E}[\theta \mid y] = w \cdot \frac{\alpha}{\alpha + \beta} + (1 - w) \cdot \frac{y}{n}$$

## 1.5 Credible Intervals and HPD Regions

**Definition 1.5.1.** An interval  $(a, b)$  is a  $(1 - \alpha)$  credible interval if

$$\mathbb{P}(a < \theta < b \mid y) = 1 - \alpha$$

**Definition 1.5.2.** A highest posterior density (HPD) region  $A$  satisfies:

- $\mathbb{P}(\theta \in A \mid y) = 1 - \alpha$
- $p(\theta_1 \mid y) > p(\theta_2 \mid y)$  for  $\theta_1 \in A$ ,  $\theta_2 \notin A$

## 1.6 Posterior Predictive Distribution

**Proposition 1.6.1.** *For a new observation  $Z$ ,*

$$p(Z = 1 \mid y) = \mathbb{E}[\theta \mid y]$$

*Proof.*

$$p(Z = 1 \mid y) = \int p(Z = 1 \mid \theta) p(\theta \mid y) d\theta = \int \theta p(\theta \mid y) d\theta = \mathbb{E}[\theta \mid y]$$

□

## 1.7 Exponential Family

**Definition 1.7.1.** A distribution belongs to the exponential family if it can be written as:

$$p(y | \theta) = f(y)g(\theta) \exp\{\phi(\theta)u(y)\}$$

**Remark 1.7.1.** Many common distributions (Normal, Binomial, Poisson) belong to this family, which enables conjugate prior constructions.

## 1.8 Prior Specification

### 1.8.1 Subjective Priors

Constructed from expert knowledge.

### 1.8.2 Non-informative Priors

Designed to have minimal influence on the posterior.

**Example 1.8.1.** A flat prior:

$$\pi(\theta) \propto 1$$

### 1.8.3 Jeffreys Prior

**Definition 1.8.1.**

$$\pi(\theta) \propto \sqrt{I(\theta)}, \quad I(\theta) = -\mathbb{E} \left[ \frac{\partial^2}{\partial \theta^2} \log p(y | \theta) \right]$$

**Remark 1.8.1.** Jeffreys priors are invariant under reparameterization.

## 1.9 Poisson-Gamma Model

### 1.9.1 Model

$$Y_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)$$

Likelihood:

$$L(\theta) \propto \theta^{\sum y_i} e^{-n\theta}$$

### 1.9.2 Prior

$$\theta \sim \text{Gamma}(a, b)$$

### 1.9.3 Posterior

**Proposition 1.9.1.**

$$\theta | y \sim \text{Gamma}(a + \sum y_i, b + n)$$

## 1.10 Preview: Computational Methods

In many models, posterior distributions cannot be computed analytically. This motivates:

- Monte Carlo integration
- Importance sampling
- Markov Chain Monte Carlo (MCMC)

These methods will be developed in later chapters.

## 1.11 Common Distributions and Identities

### 1.11.1 Beta and Gamma Functions

**Definition 1.11.1.** The Gamma function is defined as:

$$\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx$$

and satisfies:

$$\Gamma(n) = (n-1)! \quad \text{for integer } n$$

**Definition 1.11.2.** The Beta distribution has density:

$$f(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1}$$

**Proposition 1.11.1.** For  $\theta \sim \text{Beta}(a, b)$ :

$$\mathbb{E}[\theta] = \frac{a}{a+b}, \quad \text{Var}(\theta) = \frac{ab}{(a+b)^2(a+b+1)}$$

### 1.11.2 Gamma Distribution

**Definition 1.11.3.** A Gamma distribution with shape  $a$  and rate  $b$ :

$$f(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}$$

**Proposition 1.11.2.**

$$\mathbb{E}[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}$$

**Remark 1.11.1.** The Gamma distribution is conjugate to the Poisson likelihood.

## 1.12 Sufficiency and Likelihood Structure

**Definition 1.12.1.** A statistic  $T(y)$  is sufficient for  $\theta$  if the likelihood can be written as:

$$p(y \mid \theta) = g(T(y), \theta)h(y)$$

**Proposition 1.12.1.** For:

- *Binomial model: sufficient statistic is  $y = \sum y_i$*
- *Poisson model: sufficient statistic is  $\sum y_i$*

**Remark 1.12.1.** This reduction is critical in Bayesian updating since posterior distributions depend only on sufficient statistics.

## 1.13 Posterior as a Weighted Average

A key Bayesian insight is that posterior estimates combine prior and data:

**Proposition 1.13.1.** For the Beta-Binomial model:

$$\mathbb{E}[\theta \mid y] = \frac{a + y}{a + b + n}$$

can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{a + b}{a + b + n} \cdot \frac{a}{a + b} + \frac{n}{a + b + n} \cdot \frac{y}{n}$$

**Remark 1.13.1.** This shows:

- Prior contributes as pseudo-observations
- Data influence increases with  $n$

## 1.14 Bias, Variance, and MSE

**Definition 1.14.1.** Bias of an estimator  $\hat{\theta}$ :

$$\text{Bias} = \mathbb{E}[\hat{\theta}] - \theta$$

**Definition 1.14.2.** Mean Squared Error:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2$$

**Remark 1.14.1.** Bayesian estimators are often biased but achieve lower MSE through shrinkage.

## 1.15 Poisson-Gamma Model (Detailed)

### 1.15.1 Likelihood

$$L(\theta) = \prod_{i=1}^n \frac{\theta^{y_i} e^{-\theta}}{y_i!} \propto \theta^{\sum y_i} e^{-n\theta}$$

### 1.15.2 Posterior Derivation

*Proof.* With prior  $\theta \sim \text{Gamma}(a, b)$ :

$$\begin{aligned} p(\theta \mid y) &\propto \theta^{\sum y_i} e^{-n\theta} \cdot \theta^{a-1} e^{-b\theta} \\ &= \theta^{a+\sum y_i-1} e^{-(b+n)\theta} \end{aligned}$$

which is  $\text{Gamma}(a + \sum y_i, b + n)$ . □

### 1.15.3 Posterior Mean Interpretation

$$\mathbb{E}[\theta \mid y] = \frac{b}{b+n} \cdot \frac{a}{b} + \frac{n}{b+n} \cdot \frac{\sum y_i}{n}$$

**Remark 1.15.1.** Again, this is a weighted average of prior mean and sample mean.

## 1.16 Posterior Predictive Distribution

**Definition 1.16.1.** The posterior predictive distribution is:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) d\theta$$

**Remark 1.16.1.** This integrates out parameter uncertainty, a key advantage over plug-in estimators.

## 1.17 Exponential Family and Conjugacy

**Definition 1.17.1.** A one-parameter exponential family has the form:

$$p(y \mid \phi) = h(y) c(\phi) \exp\{\phi t(y)\}$$

**Proposition 1.17.1.** For exponential families:

- Sufficient statistics appear naturally as  $t(y)$
- Conjugate priors exist due to exponential structure

**Remark 1.17.1.** Examples:

- Binomial  $\leftrightarrow$  Beta
- Poisson  $\leftrightarrow$  Gamma
- Normal  $\leftrightarrow$  Normal / Inverse-Gamma

## 1.18 Interpretation of Bayesian Updating

**Remark 1.18.1.** Bayesian inference provides:

- Full uncertainty quantification (posterior distribution)
- Direct probabilistic statements about parameters
- Natural predictive distributions

**Example 1.18.1.** In rare-event estimation, even if  $y = 0$ , the posterior remains well-defined:

$$\theta \mid y \sim \text{Beta}(a, b + n)$$

which avoids degenerate estimates like  $\hat{\theta} = 0$ .



# Chapter 2

## The Binomial Model

### 2.1 Model Setup

**Definition 2.1.1.** Let  $Y \mid \theta \sim \text{Binomial}(n, \theta)$  with probability mass function:

$$p(y \mid \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}, \quad \theta \in [0, 1]$$

**Remark 2.1.1.** If  $Y_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ , then  $Y = \sum_{i=1}^n Y_i \sim \text{Binomial}(n, \theta)$ . The sufficient statistic is  $y = \sum Y_i$ .

### 2.2 Likelihood and Maximum Likelihood Estimation

Ignoring constants:

$$L(\theta) \propto \theta^y (1 - \theta)^{n-y}$$

**Proposition 2.2.1.** *The log-likelihood is:*

$$\ell(\theta) = y \log \theta + (n - y) \log(1 - \theta)$$

**Proposition 2.2.2.** *The MLE for  $\theta$  is:*

$$\hat{\theta} = \frac{y}{n}$$

*Proof.* Differentiate:

$$\frac{d\ell}{d\theta} = \frac{y}{\theta} - \frac{n-y}{1-\theta}$$

Set equal to zero:

$$\frac{y}{\theta} = \frac{n-y}{1-\theta} \Rightarrow \theta = \frac{y}{n}$$

□

## 2.3 Beta Prior

**Definition 2.3.1.** Assume prior:

$$\theta \sim \text{Beta}(a, b), \quad \pi(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$$

**Remark 2.3.1.** The parameters  $(a, b)$  can be interpreted as prior pseudo-counts:

$$a - 1 \text{ successes, } b - 1 \text{ failures}$$

## 2.4 Posterior Distribution

**Theorem 2.4.1.** If  $\theta \sim \text{Beta}(a, b)$  and  $Y \mid \theta \sim \text{Binomial}(n, \theta)$ , then:

$$\theta \mid y \sim \text{Beta}(a + y, b + n - y)$$

*Proof.*

$$\begin{aligned} p(\theta \mid y) &\propto \theta^y (1 - \theta)^{n-y} \cdot \theta^{a-1} (1 - \theta)^{b-1} \\ &= \theta^{a+y-1} (1 - \theta)^{b+n-y-1} \end{aligned}$$

□

## 2.5 Posterior Summaries

### 2.5.1 Mean

$$\mathbb{E}[\theta \mid y] = \frac{a + y}{a + b + n}$$

**Remark 2.5.1.** This can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{a + b}{a + b + n} \cdot \frac{a}{a + b} + \frac{n}{a + b + n} \cdot \frac{y}{n}$$

### 2.5.2 Mode

$$\text{Mode}(\theta \mid y) = \frac{a + y - 1}{a + b + n - 2}$$

### 2.5.3 Variance

$$\text{Var}(\theta \mid y) = \frac{(a + y)(b + n - y)}{(a + b + n)^2(a + b + n + 1)}$$

## 2.6 Posterior Predictive Distribution

**Definition 2.6.1.** For a new observation  $\tilde{Y}$ :

$$p(\tilde{Y} = 1 \mid y) = \int \theta p(\theta \mid y) d\theta$$

**Proposition 2.6.1.**

$$p(\tilde{Y} = 1 \mid y) = \mathbb{E}[\theta \mid y] = \frac{a + y}{a + b + n}$$

## 2.7 Credible Intervals

**Definition 2.7.1.** An interval  $(l, u)$  is a  $(1 - \alpha)$  credible interval if:

$$\mathbb{P}(l < \theta < u \mid y) = 1 - \alpha$$

## 2.8 Highest Posterior Density Regions

**Definition 2.8.1.** A set  $A$  is a HPD region if:

- $\mathbb{P}(\theta \in A \mid y) = 1 - \alpha$
- $p(\theta_1 \mid y) > p(\theta_2 \mid y)$  for  $\theta_1 \in A$ ,  $\theta_2 \notin A$

## 2.9 Shrinkage Interpretation

**Proposition 2.9.1.** *The posterior mean can be written as:*

$$\mathbb{E}[\theta \mid y] = w \cdot \frac{a}{a + b} + (1 - w) \cdot \frac{y}{n}$$

where:

$$w = \frac{a + b}{a + b + n}$$

**Remark 2.9.1.** • Small  $n$ : prior dominates

- Large  $n$ : data dominates

## 2.10 Rare Event Example

**Example 2.10.1.** Suppose:

$$n = 20, \quad y = 0, \quad \theta \sim \text{Beta}(2, 20)$$

Then:

$$\theta \mid y \sim \text{Beta}(2, 40)$$

**Remark 2.10.1.** The MLE gives  $\hat{\theta} = 0$ , while the Bayesian estimator remains strictly positive.

## 2.11 Frequentist vs Bayesian Interpretation

**Remark 2.11.1.** Frequentist interval:

$$\mathbb{P}(l(Y) < \theta < u(Y) \mid \theta) = 1 - \alpha$$

Bayesian interval:

$$\mathbb{P}(l < \theta < u \mid y) = 1 - \alpha$$

# Chapter 3

## Poisson Model and Exponential Families

### 3.1 Poisson Model

**Definition 3.1.1.** A random variable  $Y$  follows a Poisson distribution with rate  $\theta > 0$  if:

$$Y \mid \theta \sim \text{Poisson}(\theta), \quad p(y \mid \theta) = \frac{\theta^y e^{-\theta}}{y!}, \quad y \in \mathbb{N}$$

**Proposition 3.1.1.** For  $Y \sim \text{Poisson}(\theta)$ :

$$\mathbb{E}[Y] = \theta, \quad \text{Var}(Y) = \theta$$

**Remark 3.1.1.** The equality of mean and variance is a defining feature of the Poisson model. Large deviations between sample mean and variance indicate model misspecification.

### 3.2 Likelihood and Sufficiency

Let  $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)$ . The joint likelihood is:

$$p(y_1, \dots, y_n \mid \theta) = \prod_{i=1}^n \frac{\theta^{y_i} e^{-\theta}}{y_i!} = c(y) \theta^{\sum y_i} e^{-n\theta}$$

**Proposition 3.2.1.** The sufficient statistic is:

$$T(y) = \sum_{i=1}^n y_i$$

**Remark 3.2.1.** All information about  $\theta$  is contained in the total count  $\sum y_i$ .

### 3.3 Maximum Likelihood Estimation

**Proposition 3.3.1.** *The MLE for  $\theta$  is:*

$$\hat{\theta} = \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

*Proof.* Log-likelihood:

$$\ell(\theta) = \sum y_i \log \theta - n\theta$$

Differentiate:

$$\frac{d\ell}{d\theta} = \frac{\sum y_i}{\theta} - n$$

Set to zero:

$$\theta = \frac{1}{n} \sum y_i$$

□

### 3.4 Gamma Prior

**Definition 3.4.1.** Assume prior:

$$\theta \sim \text{Gamma}(a, b), \quad \pi(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}$$

**Proposition 3.4.1.**

$$\mathbb{E}[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}$$

### 3.5 Posterior Distribution

**Theorem 3.5.1.** *If  $\theta \sim \text{Gamma}(a, b)$  and  $Y_i \stackrel{iid}{\sim} \text{Poisson}(\theta)$ , then:*

$$\theta \mid y \sim \text{Gamma}\left(a + \sum y_i, b + n\right)$$

*Proof.*

$$p(\theta \mid y) \propto \theta^{\sum y_i} e^{-n\theta} \cdot \theta^{a-1} e^{-b\theta}$$

$$= \theta^{a+\sum y_i-1} e^{-(b+n)\theta}$$

□

## 3.6 Posterior Summaries

### 3.6.1 Mean

$$\mathbb{E}[\theta \mid y] = \frac{a + \sum y_i}{b + n}$$

**Remark 3.6.1.** This can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{b}{b + n} \cdot \frac{a}{b} + \frac{n}{b + n} \cdot \frac{\sum y_i}{n}$$

### 3.6.2 Variance

$$\text{Var}(\theta \mid y) = \frac{a + \sum y_i}{(b + n)^2}$$

## 3.7 Posterior Predictive Distribution

**Definition 3.7.1.** For a new observation  $\tilde{Y}$ :

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) d\theta$$

**Remark 3.7.1.** This results in a negative binomial distribution, reflecting overdispersion relative to the Poisson model.

## 3.8 Example: Birth Rate Comparison

**Example 3.8.1.** Suppose:

- Group 1:  $n_1 = 111$ ,  $\sum y_i = 217$
- Group 2:  $n_2 = 44$ ,  $\sum y_i = 66$
- Prior:  $\theta \sim \text{Gamma}(2, 1)$

Posteriors:

$$\theta_1 \mid y \sim \text{Gamma}(219, 112), \quad \theta_2 \mid y \sim \text{Gamma}(68, 45)$$

## 3.9 Exponential Family Representation

**Definition 3.9.1.** A distribution belongs to the exponential family if:

$$p(y \mid \phi) = h(y)c(\phi) \exp\{\phi t(y)\}$$

### 3.9.1 Poisson as an Exponential Family

**Proposition 3.9.1.** *The Poisson distribution can be written as:*

$$p(y \mid \theta) = \frac{1}{y!} \exp\{y \log \theta - \theta\}$$

**Remark 3.9.1.** This corresponds to:

$$t(y) = y, \quad \phi = \log \theta$$

## 3.10 General Exponential Family Structure

**Definition 3.10.1.** A one-parameter exponential family has the form:

$$p(y \mid \phi) = h(y)c(\phi) \exp\{\phi t(y)\}$$

**Proposition 3.10.1.** *For exponential families:*

- $t(y)$  is a sufficient statistic
- Conjugate priors exist due to exponential structure

**Remark 3.10.1.** Common examples:

- Binomial  $\leftrightarrow$  Beta
- Poisson  $\leftrightarrow$  Gamma
- Normal  $\leftrightarrow$  Normal / Inverse-Gamma

## 3.11 Interpretation and Model Limitations

**Remark 3.11.1.** The Poisson model assumes:

- independence of events
- constant rate  $\theta$
- equal mean and variance

**Remark 3.11.2.** If variance exceeds the mean, a negative binomial model may be more appropriate.



# Chapter 4

## Monte Carlo Methods

### 4.1 Motivation

**Remark 4.1.1.** In many Bayesian models, posterior quantities cannot be computed analytically. Instead, we approximate expectations using simulation methods.

**Definition 4.1.1.** We are often interested in computing:

$$\mathbb{E}[f(\theta)] = \int f(\theta)p(\theta) d\theta$$

**Remark 4.1.2.** Monte Carlo methods approximate this integral using random samples.

### 4.2 Monte Carlo Estimation

**Definition 4.2.1.** Let  $\theta^{(1)}, \dots, \theta^{(S)} \sim p(\theta)$  be i.i.d. samples. Then:

$$\hat{\mu} = \frac{1}{S} \sum_{s=1}^S f(\theta^{(s)})$$

**Proposition 4.2.1.** *By the Law of Large Numbers:*

$$\hat{\mu} \rightarrow \mathbb{E}[f(\theta)] \quad \text{as } S \rightarrow \infty$$

#### 4.2.1 Variance and Error

**Definition 4.2.2.** Monte Carlo variance:

$$\widehat{\text{Var}} = \frac{1}{S-1} \sum_{s=1}^S \left( f(\theta^{(s)}) - \hat{\mu} \right)^2$$

**Definition 4.2.3.** Monte Carlo standard error:

$$\text{MCSE} = \sqrt{\frac{\widehat{\text{Var}}}{S}}$$

**Definition 4.2.4.** Monte Carlo interval:

$$\hat{\mu} \pm 2 \cdot \text{MCSE}$$

## 4.3 Importance Sampling

**Remark 4.3.1.** Direct sampling from  $p(\theta)$  is often difficult. Instead, we sample from an easier distribution  $g(\theta)$ .

**Definition 4.3.1.** Rewrite the expectation:

$$\mathbb{E}[f(\theta)] = \int f(\theta) \frac{p(\theta)}{g(\theta)} g(\theta) d\theta$$

**Definition 4.3.2.** Define weights:

$$w(\theta) = \frac{p(\theta)}{g(\theta)}$$

### 4.3.1 Estimators

- Unnormalized:

$$\hat{\mu}^* = \frac{1}{S} \sum_{s=1}^S f(\theta^{(s)}) w^*(\theta^{(s)})$$

- Normalized:

$$\hat{\mu} = \sum_{s=1}^S f(\theta^{(s)}) w(\theta^{(s)})$$

where:

$$w(\theta^{(s)}) = \frac{w^*(\theta^{(s)})}{\sum_{j=1}^S w^*(\theta^{(j)})}$$

### 4.3.2 Effective Sample Size

**Definition 4.3.3.** Effective sample size (ESS):

$$\text{ESS} \approx \frac{S}{1 + \text{Var}(w^*)}$$

**Remark 4.3.2.** High variance in weights reduces efficiency.

## 4.4 Accept-Reject Sampling

**Definition 4.4.1.** Suppose we want to sample from  $p(x)$  using proposal  $g(x)$  such that:

$$p(x) \leq M g(x)$$

### 4.4.1 Algorithm

1. Sample  $X \sim g(x)$
2. Sample  $U \sim \text{Uniform}(0, 1)$
3. Accept if:

$$U \leq \frac{p(X)}{Mg(X)}$$

4. Otherwise reject and repeat

**Remark 4.4.1.** Efficiency depends on choosing  $M$  as small as possible.

## 4.5 Limitations of Direct Methods

**Remark 4.5.1.** Monte Carlo methods require:

- ability to sample from target distribution
- tractable density evaluation

**Remark 4.5.2.** When this is not possible, we use Markov Chain Monte Carlo (MCMC).

## 4.6 Markov Chains

**Definition 4.6.1.** A Markov chain satisfies:

$$P(X_{n+1} \mid X_n, \dots, X_0) = P(X_{n+1} \mid X_n)$$

**Definition 4.6.2.** A stationary distribution  $\pi$  satisfies:

$$\pi = \pi P$$

**Remark 4.6.1.** We construct chains whose stationary distribution is the posterior  $p(\theta|y)$ .

## 4.7 Markov Chain Monte Carlo (MCMC)

**Definition 4.7.1.** MCMC methods generate dependent samples  $\theta^{(1)}, \dots, \theta^{(S)}$  such that:

$$\theta^{(s)} \sim p(\theta \mid y) \quad \text{as } s \rightarrow \infty$$

**Proposition 4.7.1.** *Ergodic theorem:*

$$\frac{1}{S} \sum_{s=1}^S f(\theta^{(s)}) \rightarrow \mathbb{E}[f(\theta \mid y)]$$

**Remark 4.7.1.** Samples are no longer independent, which affects variance.

## 4.8 Metropolis-Hastings Algorithm

**Definition 4.8.1.** Given proposal  $q(\theta^* | \theta)$ , define acceptance probability:

$$\alpha(\theta, \theta^*) = \min \left\{ \frac{p(\theta^* | y)q(\theta | \theta^*)}{p(\theta | y)q(\theta^* | \theta)}, 1 \right\}$$

### 4.8.1 Algorithm

1. Initialize  $\theta^{(0)}$
2. Propose  $\theta^* \sim q(\cdot | \theta^{(t-1)})$
3. Accept with probability  $\alpha$
4. Otherwise keep  $\theta^{(t)} = \theta^{(t-1)}$

**Remark 4.8.1.** If  $q$  is symmetric, the ratio simplifies.

## 4.9 Random Walk Metropolis

**Definition 4.9.1.** Proposal:

$$\theta^* = \theta^{(t-1)} + \epsilon$$

**Remark 4.9.1.** Common choices for  $\epsilon$ :

- Normal
- Student-t
- Uniform

## 4.10 Gibbs Sampling

**Definition 4.10.1.** Instead of sampling jointly, we sample from conditionals:

$$\theta_1 \sim p(\theta_1 | \theta_2, y), \quad \theta_2 \sim p(\theta_2 | \theta_1, y)$$

**Remark 4.10.1.** This is especially useful in high-dimensional problems.

## 4.11 Burn-in and Convergence

**Definition 4.11.1.** Burn-in: discard first  $B$  samples:

$$\{\theta^{(B+1)}, \dots, \theta^{(B+S)}\}$$

**Remark 4.11.1.** Early samples depend on initialization and may not represent the target distribution.

## 4.12 Autocorrelation and Efficiency

**Definition 4.12.1.** Autocorrelation at lag  $t$ :

$$\rho_t = \frac{\sum_{s=1}^{S-t} (\theta_s - \bar{\theta})(\theta_{s+t} - \bar{\theta})}{\sum_{s=1}^S (\theta_s - \bar{\theta})^2}$$

**Remark 4.12.1.** High autocorrelation implies slow mixing and inefficient sampling.

## 4.13 Diagnostics

- Trace plots
- Gelman-Rubin statistic
- Geweke test
- Effective sample size

**Remark 4.13.1.** No diagnostic guarantees convergence—only detects problems.

## 4.14 Summary

- Monte Carlo approximates integrals using random sampling
- Importance sampling improves flexibility
- Accept-reject enables indirect sampling
- MCMC handles complex posterior distributions
- Efficiency depends on mixing and autocorrelation

# Chapter 5

## Normal Model

### 5.1 The Normal Sampling Model

**Definition 5.1.1.** A random variable  $Y$  follows a normal distribution if:

$$Y \mid \theta, \sigma^2 \sim \mathcal{N}(\theta, \sigma^2)$$

with density:

$$p(y \mid \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right)$$

**Proposition 5.1.1.** For  $Y \sim \mathcal{N}(\theta, \sigma^2)$ :

$$\mathbb{E}[Y] = \theta, \quad \text{Var}(Y) = \sigma^2$$

**Remark 5.1.1.** The normal distribution is symmetric with mean = median = mode, and approximately 95% of mass lies within  $\theta \pm 1.96\sigma$ .

### 5.2 Likelihood Function

Let  $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2)$ . Then:

$$p(y \mid \theta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \theta)^2\right)$$

**Remark 5.2.1.** Inference can be separated into:

- inference on  $\theta$  given  $\sigma^2$
- inference on  $\sigma^2$

### 5.3 Known Variance: Conjugate Analysis

#### 5.3.1 Prior

**Definition 5.3.1.** Assume:

$$\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$$

### 5.3.2 Posterior

**Theorem 5.3.1.** If  $Y_i \stackrel{iid}{\sim} \mathcal{N}(\theta, \sigma^2)$  and  $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$ , then:

$$\theta \mid y \sim \mathcal{N}(\mu_n, \tau_n^2)$$

where:

$$\begin{aligned} \frac{1}{\tau_n^2} &= \frac{1}{\tau_0^2} + \frac{n}{\sigma^2} \\ \mu_n &= \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \end{aligned}$$

*Proof.* Posterior:

$$p(\theta \mid y) \propto p(\theta) \cdot p(y \mid \theta)$$

Combine exponent terms:

$$-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2 - \frac{1}{2\sigma^2} \sum (y_i - \theta)^2$$

Expand and collect quadratic terms in  $\theta$ , yielding a normal kernel. □

### 5.3.3 Interpretation

**Remark 5.3.1.** The posterior mean is a weighted average:

$$\mu_n = \frac{\kappa_0}{\kappa_0 + n} \mu_0 + \frac{n}{\kappa_0 + n} \bar{y}$$

where  $\kappa_0 = \sigma^2 / \tau_0^2$ .

**Remark 5.3.2.** Precision (inverse variance) is additive:

$$\text{precision}_{\text{posterior}} = \text{precision}_{\text{prior}} + \text{precision}_{\text{data}}$$

## 5.4 Unknown Variance: Joint Inference

**Remark 5.4.1.** When  $\sigma^2$  is unknown, we introduce a prior on variance.

### 5.4.1 Prior Specification

**Definition 5.4.1.** Use:

$$\begin{aligned} \theta \mid \sigma^2 &\sim \mathcal{N}(\mu_0, \sigma^2 / \kappa_0) \\ \sigma^2 &\sim \text{Inv-Gamma} \left( \frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2} \right) \end{aligned}$$

### 5.4.2 Posterior Updates

**Proposition 5.4.1.** *Posterior parameters:*

$$\begin{aligned}\kappa_n &= \kappa_0 + n \\ \mu_n &= \frac{\kappa_0 \mu_0 + n \bar{y}}{\kappa_n} \\ \nu_n &= \nu_0 + n \\ \sigma_n^2 &= \frac{1}{\nu_n} \left[ \nu_0 \sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_n} (\bar{y} - \mu_0)^2 \right]\end{aligned}$$

**Remark 5.4.2.** This is often summarized as: “posterior sum of squares = prior sum + data sum + correction term.”

## 5.5 Conditional Posteriors

**Proposition 5.5.1.**

$$\begin{aligned}\theta \mid \sigma^2, y &\sim \mathcal{N}(\mu_n, \sigma^2 / \kappa_n) \\ \sigma^2 \mid y &\sim \text{Inv-Gamma} \left( \frac{\nu_n}{2}, \frac{\nu_n \sigma_n^2}{2} \right)\end{aligned}$$

## 5.6 Marginal Posterior for the Mean

**Theorem 5.6.1.** *The marginal posterior of  $\theta$  is:*

$$\frac{\theta - \mu_n}{\sigma_n / \sqrt{\kappa_n}} \sim t_{\nu_n}$$

**Remark 5.6.1.** The  $t$ -distribution arises from integrating out uncertainty in  $\sigma^2$ .

## 5.7 Example: Midge Wing Length

**Example 5.7.1.** Data:

$$y = (1.64, 1.70, 1.72, 1.74, 1.82, 1.82, 1.82, 1.90, 2.08)$$

Summary:

$$\bar{y} = 1.804, \quad s^2 \approx 0.017$$

Using prior  $\mu_0 = 1.9$ ,  $\kappa_0 = 1$ , we obtain:

$$\mu_n \approx 1.814, \quad \sigma_n^2 \approx 0.015$$



## 5.8 Sampling from the Posterior

**Remark 5.8.1.** We can sample from the joint posterior using:

1. Sample  $\sigma^2 \sim \text{Inv-Gamma}$
2. Sample  $\theta \sim \mathcal{N}$

## 5.9 Bias and Mean Squared Error

**Definition 5.9.1.** Bias:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

**Definition 5.9.2.** Mean Squared Error:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2$$

**Remark 5.9.1.** Bayesian estimators are typically biased but can have lower MSE due to shrinkage.

## 5.10 Exponential Family Representation

**Proposition 5.10.1.** *The normal distribution belongs to the exponential family:*

$$p(y \mid \theta, \sigma^2) = \exp \left( \frac{\theta}{\sigma^2} y - \frac{1}{2\sigma^2} y^2 - A(\theta, \sigma^2) \right)$$

**Remark 5.10.1.** Sufficient statistics:

$$t(y) = (y, y^2)$$

## 5.11 Summary

- Normal model is central to Bayesian data analysis
- Conjugate priors yield closed-form posteriors
- Posterior mean is a weighted average of prior and data
- Unknown variance leads to  $t$ -distributed marginal posterior
- Joint inference handled via conditional distributions

# Chapter 6

## Multinomial Model, Bayes Factors, and Indirect Methods

### 6.1 Multinomial Model

#### 6.1.1 Model Setup

**Definition 6.1.1.** Let:

$$x = (x_1, \dots, x_k), \quad \sum_{j=1}^k x_j = n$$

Then the multinomial model is:

$$p(x \mid \theta) = \prod_{j=1}^k \theta_j^{x_j}$$

where:

$$\theta = (\theta_1, \dots, \theta_k), \quad \sum_{j=1}^k \theta_j = 1$$

**Remark 6.1.1.** This generalizes the binomial model to  $k$  categories.

#### 6.1.2 Dirichlet Prior

**Definition 6.1.2.** The conjugate prior is:

$$\theta \sim \text{Dirichlet}(\alpha)$$

with density:

$$\pi(\theta) = \frac{1}{B(\alpha)} \prod_{j=1}^k \theta_j^{\alpha_j - 1}$$

**Remark 6.1.2.** •  $\alpha_j$  acts like prior counts

- $\alpha_j = 1$  gives a uniform prior
- when  $k = 2$ , Dirichlet reduces to Beta

### 6.1.3 Posterior

**Theorem 6.1.1.** *Given data  $x$ , the posterior is:*

$$\theta \mid x \sim \text{Dirichlet}(\alpha + x)$$

*Proof.*

$$p(\theta \mid x) \propto \prod_{j=1}^k \theta_j^{x_j} \cdot \prod_{j=1}^k \theta_j^{\alpha_j-1} = \prod_{j=1}^k \theta_j^{x_j+\alpha_j-1}$$

□

### 6.1.4 Interpretation

**Remark 6.1.3.** Posterior = prior counts + observed counts:

$$\alpha_j \rightarrow \alpha_j + x_j$$

**Remark 6.1.4.** Dirichlet provides a natural Bayesian smoothing for categorical probabilities.

## 6.2 Bayes Factors

### 6.2.1 Definition

**Definition 6.2.1.** Given two models  $M_1$  and  $M_2$ :

$$\text{BF} = \frac{p(M_1 \mid x)}{p(M_2 \mid x)} \cdot \frac{p(M_2)}{p(M_1)} = \frac{p(x \mid M_1)}{p(x \mid M_2)}$$

**Remark 6.2.1.** Bayes factor compares how well each model explains the data.

### 6.2.2 Marginal Likelihood

**Definition 6.2.2.**

$$p(x \mid M) = \int p(x \mid \theta, M) \pi(\theta \mid M) d\theta$$

**Remark 6.2.2.** This integrates out parameter uncertainty.

### 6.2.3 Interpretation

**Remark 6.2.3.** Typical interpretation scales (Jeffreys):

- $1 \leq BF < 3$ : weak evidence
- $3 \leq BF < 10$ : substantial
- $10 \leq BF < 100$ : strong
- $BF \geq 100$ : decisive

## 6.2.4 Important Caveats

**Remark 6.2.4.** • Bayes factors require **proper priors**

- If prior is improper  $\Rightarrow$  marginal likelihood undefined
- Sensitive to prior specification

## 6.2.5 Frequentist vs Bayesian Testing

**Remark 6.2.5.** • Frequentist: reject/not reject (no strength)

- Bayesian: provides strength of evidence via BF

# 6.3 Indirect Methods (Monte Carlo)

## 6.3.1 Motivation

**Remark 6.3.1.** We often need to compute:

$$\mathbb{E}[f(\theta)] = \int f(\theta)p(\theta) d\theta$$

but cannot evaluate the integral analytically.

# 6.4 Importance Sampling

## 6.4.1 Core Idea

**Definition 6.4.1.** Rewrite expectation using proposal  $g(x)$ :

$$\mathbb{E}[h(x)] = \int h(x)f(x) dx = \int h(x)\frac{f(x)}{g(x)}g(x) dx$$

## 6.4.2 Estimator

**Definition 6.4.2.** Unnormalized:

$$\hat{\mu}^* = \frac{1}{n} \sum_{i=1}^n h(X_i) \frac{f(X_i)}{g(X_i)}$$

Normalized:

$$\hat{\mu} = \sum_{i=1}^n h(X_i)w_i$$

where:

$$w_i = \frac{f(X_i)/g(X_i)}{\sum_j f(X_j)/g(X_j)}$$

### 6.4.3 Effective Sample Size

**Definition 6.4.3.**

$$\text{ESS} = \frac{n}{1 + \widehat{\text{Var}}(w^*)}$$

**Remark 6.4.1.** High variance weights  $\Rightarrow$  low efficiency.

## 6.5 Accept-Reject Sampling

### 6.5.1 Algorithm

- 
1. Choose proposal  $g(x)$  and constant  $M$  such that:

$$f(x) \leq Mg(x)$$

2. Sample  $X \sim g(x)$
3. Sample  $U \sim \text{Uniform}(0, 1)$
4. Accept if:

$$U \leq \frac{f(X)}{Mg(X)}$$


---

### 6.5.2 Remarks

- Smaller  $M \Rightarrow$  higher efficiency
- $g$  must be easy to sample

## 6.6 Monte Carlo Approximation

**Definition 6.6.1.** Given samples  $\theta^{(1)}, \dots, \theta^{(S)}$ :

$$\hat{\theta} = \frac{1}{S} \sum_{s=1}^S \theta^{(s)}$$

**Definition 6.6.2.** Monte Carlo variance:

$$\widehat{\text{Var}} = \frac{1}{S-1} \sum (\theta^{(s)} - \bar{\theta})^2$$

**Remark 6.6.1.** Accuracy improves with  $S$  by law of large numbers.

## 6.7 Summary

- Multinomial + Dirichlet gives clean conjugate updates
- Bayes factors compare models via marginal likelihoods
- Indirect methods approximate integrals when closed forms fail
- Importance sampling and AR are foundational tools before MCMC

# Chapter 7

## Markov Chain Monte Carlo, Gibbs Sampling, and Diagnostics

### 7.1 Motivation

**Remark 7.1.1.** In many Bayesian models, the posterior

$$p(\theta \mid y)$$

cannot be computed in closed form.

We instead approximate expectations:

$$\mathbb{E}[h(\theta) \mid y] = \int h(\theta)p(\theta \mid y) d\theta$$

using simulation methods.

### 7.2 Markov Chains

#### 7.2.1 Definition

**Definition 7.2.1.** A sequence  $\{X_t\}$  is a Markov chain if:

$$\mathbb{P}(X_{t+1} \mid X_t, \dots, X_0) = \mathbb{P}(X_{t+1} \mid X_t)$$

#### 7.2.2 Key Properties

- Irreducibility: all states communicate
- Aperiodicity: returns do not occur in fixed cycles
- Positive recurrence: expected return time is finite

**Definition 7.2.2.** A chain satisfying these properties is called ergodic.

### 7.2.3 Stationary Distribution

**Definition 7.2.3.** A distribution  $\pi$  is stationary if:

$$\pi = \pi P$$

**Remark 7.2.1.** MCMC constructs a Markov chain whose stationary distribution is the posterior:

$$\pi(\theta) = p(\theta \mid y)$$

## 7.3 Markov Chain Monte Carlo

**Definition 7.3.1.** MCMC generates dependent samples:

$$\theta^{(1)}, \theta^{(2)}, \dots$$

such that:

$$\theta^{(t)} \xrightarrow{d} p(\theta \mid y)$$

**Proposition 7.3.1.** For an ergodic chain:

$$\mathbb{E}[h(\theta) \mid y] \approx \frac{1}{T} \sum_{t=1}^T h(\theta^{(t)})$$

## 7.4 Metropolis-Hastings Algorithm

**Definition 7.4.1.** Given a proposal  $q(\theta^* \mid \theta)$ , define:

$$\alpha(\theta, \theta^*) = \min \left\{ \frac{p(\theta^* \mid y)q(\theta \mid \theta^*)}{p(\theta \mid y)q(\theta^* \mid \theta)}, 1 \right\}$$

- 
1. Initialize  $\theta^{(0)}$
  2. For  $t = 1, \dots, T$ :
    - (a) Sample  $\theta^* \sim q(\cdot \mid \theta^{(t-1)})$
    - (b) Accept with probability  $\alpha$
    - (c) Otherwise set  $\theta^{(t)} = \theta^{(t-1)}$
- 

**Remark 7.4.1.** If  $q$  is symmetric, the ratio simplifies to:

$$\alpha = \min \left\{ \frac{p(\theta^* \mid y)}{p(\theta \mid y)}, 1 \right\}$$



## 7.5 Random Walk Metropolis

**Definition 7.5.1.** A common proposal is:

$$\theta^* = \theta^{(t-1)} + \epsilon$$

**Remark 7.5.1.** The choice of proposal variance determines:

- step size
- acceptance rate
- efficiency

## 7.6 Gibbs Sampling

### 7.6.1 Definition

**Definition 7.6.1.** Let  $\theta = (\theta_1, \dots, \theta_k)$ . Gibbs sampling iteratively samples:

$$\begin{aligned}\theta_1^{(t)} &\sim p(\theta_1 \mid \theta_2^{(t-1)}, \dots, \theta_k^{(t-1)}, y) \\ \theta_2^{(t)} &\sim p(\theta_2 \mid \theta_1^{(t)}, \dots, \theta_k^{(t-1)}, y) \\ &\vdots \\ \theta_k^{(t)} &\sim p(\theta_k \mid \theta_1^{(t)}, \dots, \theta_{k-1}^{(t)}, y)\end{aligned}$$

**Remark 7.6.1.** Gibbs sampling avoids rejection by sampling directly from full conditional distributions.

### 7.6.2 Example: Normal Model

- $\theta \mid \sigma^2, y \sim \mathcal{N}(\mu_n, \sigma^2/\kappa_n)$
- $\sigma^2 \mid \theta, y \sim \text{Inv-Gamma}(\cdot)$

**Remark 7.6.2.** Alternating these steps produces samples from the joint posterior.

## 7.7 Burn-in

**Definition 7.7.1.** Discard the first  $B$  samples:

$$\{\theta^{(B+1)}, \dots, \theta^{(T)}\}$$

**Remark 7.7.1.** Initial samples depend on the starting value and may not represent the posterior.

## 7.8 Autocorrelation

**Definition 7.8.1.** Lag- $k$  autocorrelation:

$$\rho_k = \frac{\sum_{t=1}^{T-k} (\theta^{(t)} - \bar{\theta})(\theta^{(t+k)} - \bar{\theta})}{\sum_{t=1}^T (\theta^{(t)} - \bar{\theta})^2}$$

**Remark 7.8.1.** High autocorrelation implies slow exploration of the parameter space.

## 7.9 Effective Sample Size

**Definition 7.9.1.**

$$\text{ESS} = \frac{T}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

**Remark 7.9.1.** ESS measures the number of independent samples equivalent to the correlated chain.

## 7.10 Trace Plots

**Remark 7.10.1.** Plotting  $\theta^{(t)}$  against  $t$  provides a visual diagnostic:

- stable band indicates convergence
- trends or clustering indicate poor mixing

## 7.11 Formal Diagnostics

- Geweke diagnostic: compares early and late means
- Gelman-Rubin: compares multiple chains
- Heidelberger-Welch: tests stationarity
- Raftery-Lewis: estimates required sample size

**Remark 7.11.1.** Diagnostics can detect lack of convergence but cannot guarantee convergence.

## 7.12 Mixing

**Definition 7.12.1.** Mixing describes how quickly the chain explores the parameter space.

**Remark 7.12.1.** • fast mixing: low autocorrelation

- slow mixing: high autocorrelation

## 7.13 Improving MCMC Performance

- run longer chains
- tune proposal distributions
- reparameterize the model
- block correlated variables
- introduce auxiliary variables

## 7.14 Summary

- MCMC approximates posterior expectations using dependent samples
- Metropolis-Hastings is general-purpose
- Gibbs sampling uses conditional structure
- diagnostics are essential for assessing convergence
- efficiency depends on mixing and autocorrelation

# Chapter 8

## Multivariate Normal Model

### 8.1 Multivariate Normal Distribution

**Definition 8.1.1.** A random vector  $Y \in \mathbb{R}^p$  follows a multivariate normal distribution if:

$$Y \mid \theta, \Sigma \sim \mathcal{N}_p(\theta, \Sigma)$$

with density:

$$p(y \mid \theta, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left( -\frac{1}{2} (y - \theta)^\top \Sigma^{-1} (y - \theta) \right)$$

**Proposition 8.1.1.**

$$\mathbb{E}[Y] = \theta, \quad \text{Var}(Y) = \Sigma$$

**Remark 8.1.1.** The covariance matrix  $\Sigma$  must be symmetric positive definite.

### 8.2 Sampling Model

Let  $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \mathcal{N}_p(\theta, \Sigma)$ .

**Proposition 8.2.1.** *The likelihood is:*

$$p(y \mid \theta, \Sigma) \propto |\Sigma|^{-n/2} \exp \left( -\frac{1}{2} \sum_{i=1}^n (y_i - \theta)^\top \Sigma^{-1} (y_i - \theta) \right)$$

**Remark 8.2.1.** The sufficient statistics are:

$$\bar{y} = \frac{1}{n} \sum y_i, \quad S = \sum (y_i - \bar{y})(y_i - \bar{y})^\top$$

### 8.3 Known Covariance Matrix

#### 8.3.1 Prior

**Definition 8.3.1.**

$$\theta \sim \mathcal{N}_p(\mu_0, \Lambda_0)$$

### 8.3.2 Posterior

**Theorem 8.3.1.**

$$\theta \mid y, \Sigma \sim \mathcal{N}_p(\mu_n, \Lambda_n)$$

where:

$$\begin{aligned}\Lambda_n &= \left( \Lambda_0^{-1} + n\Sigma^{-1} \right)^{-1} \\ \mu_n &= \Lambda_n \left( \Lambda_0^{-1} \mu_0 + n\Sigma^{-1} \bar{y} \right)\end{aligned}$$

**Remark 8.3.1.** Posterior precision:

$$\Lambda_n^{-1} = \Lambda_0^{-1} + n\Sigma^{-1}$$

**Remark 8.3.2.** Posterior mean is a weighted average of prior mean and sample mean.

## 8.4 Unknown Covariance Matrix

### 8.4.1 Inverse-Wishart Prior

**Definition 8.4.1.**

$$\Sigma \sim \text{Inv-Wishart}(\nu_0, S_0)$$

**Remark 8.4.1.** This is the multivariate analogue of the inverse-gamma distribution.

### 8.4.2 Joint Prior

**Definition 8.4.2.**

$$\begin{aligned}\theta \mid \Sigma &\sim \mathcal{N}_p(\mu_0, \Sigma/\kappa_0) \\ \Sigma &\sim \text{Inv-Wishart}(\nu_0, S_0)\end{aligned}$$

## 8.5 Posterior Distribution

**Proposition 8.5.1.** *Posterior parameters:*

$$\begin{aligned}\kappa_n &= \kappa_0 + n \\ \mu_n &= \frac{\kappa_0 \mu_0 + n \bar{y}}{\kappa_n} \\ \nu_n &= \nu_0 + n \\ S_n &= S_0 + \sum_{i=1}^n (y_i - \bar{y})(y_i - \bar{y})^\top + \frac{\kappa_0 n}{\kappa_n} (\bar{y} - \mu_0)(\bar{y} - \mu_0)^\top\end{aligned}$$

**Proposition 8.5.2.**

$$\begin{aligned}\theta \mid \Sigma, y &\sim \mathcal{N}_p(\mu_n, \Sigma/\kappa_n) \\ \Sigma \mid y &\sim \text{Inv-Wishart}(\nu_n, S_n)\end{aligned}$$

---

1. Initialize  $\theta^{(0)}, \Sigma^{(0)}$

2. For  $t = 1, \dots, T$ :

(a) Sample:

$$\theta^{(t)} \sim \mathcal{N}_p(\mu_n, \Sigma^{(t-1)} / \kappa_n)$$

(b) Sample:

$$\Sigma^{(t)} \sim \text{Inv-Wishart}(\nu_n, S_n(\theta^{(t)}))$$


---

## 8.6 Gibbs Sampling for MVN Model

**Remark 8.6.1.** This alternates between conditional distributions, producing samples from the joint posterior.

## 8.7 Wishart Distribution

**Definition 8.7.1.** The Wishart distribution is:

$$W \sim \text{Wishart}_p(\nu, \Sigma)$$

**Proposition 8.7.1.**

$$\mathbb{E}[W] = \nu \Sigma$$

**Remark 8.7.1.** The Wishart distribution is the distribution of a sample covariance matrix.

## 8.8 Inverse-Wishart Distribution

**Proposition 8.8.1.** If  $\Sigma \sim \text{Inv-Wishart}(\nu, S)$ :

$$\mathbb{E}[\Sigma] = \frac{S}{\nu - p - 1}$$

**Remark 8.8.1.** Defined only when  $\nu > p + 1$ .

## 8.9 Multivariate t-Distribution

**Proposition 8.9.1.** Marginalizing  $\Sigma$  yields:

$$\theta \mid y \sim t_{\nu_n - p + 1}(\mu_n, S_n)$$

**Remark 8.9.1.** The multivariate  $t$  distribution accounts for uncertainty in covariance estimation.

## 8.10 Simulation of MVN

**Proposition 8.10.1.** *If  $Z \sim \mathcal{N}_p(0, I)$  and  $\Sigma = LL^\top$ , then:*

$$Y = \theta + LZ \sim \mathcal{N}_p(\theta, \Sigma)$$

## 8.11 Summary

- Multivariate normal generalizes the normal model to vector-valued data
- Mean becomes a vector, variance becomes a covariance matrix
- Conjugate prior: Normal-Inverse-Wishart
- Posterior updates mirror univariate case using matrix algebra
- Gibbs sampling alternates between mean and covariance updates

# Chapter 9

## Group Comparisons and Hierarchical Models

### 9.1 Motivation

**Remark 9.1.1.** Many datasets are naturally grouped:

- Students within schools
- Patients within hospitals
- Firms within industries

**Remark 9.1.2.** We want to model:

- Within-group variability
- Between-group variability

### 9.2 Two-Group Comparison Model

#### 9.2.1 Model Specification

**Definition 9.2.1.** For two groups:

$$Y_{i,1} = \mu + \delta + \epsilon_{i,1}$$

$$Y_{i,2} = \mu - \delta + \epsilon_{i,2}$$

$$\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$$

**Remark 9.2.1.** •  $\mu$  = overall mean

- $\delta$  = group difference
- $\sigma^2$  = shared variance



### 9.3 Priors

**Definition 9.3.1.**

$$\begin{aligned}\mu &\sim \mathcal{N}(\mu_0, \gamma_0^2) \\ \delta &\sim \mathcal{N}(\delta_0, \tau_0^2) \\ \sigma^2 &\sim \text{Inv-Gamma}(\nu_0/2, \nu_0\sigma_0^2/2)\end{aligned}$$

### 9.4 Posterior Conditionals

#### 9.4.1 Posterior of $\mu$

**Proposition 9.4.1.**

$$\begin{aligned}\mu \mid \delta, \sigma^2, y &\sim \mathcal{N}(\mu_n, \gamma_n^2) \\ \gamma_n^2 &= \left( \frac{1}{\gamma_0^2} + \frac{n_1 + n_2}{\sigma^2} \right)^{-1} \\ \mu_n &= \gamma_n^2 \left[ \frac{\mu_0}{\gamma_0^2} + \frac{\sum(y_{i,1} - \delta)}{\sigma^2} + \frac{\sum(y_{i,2} + \delta)}{\sigma^2} \right]\end{aligned}$$

#### 9.4.2 Posterior of $\delta$

**Proposition 9.4.2.**

$$\begin{aligned}\delta \mid \mu, \sigma^2, y &\sim \mathcal{N}(\delta_n, \tau_n^2) \\ \tau_n^2 &= \left( \frac{1}{\tau_0^2} + \frac{n_1 + n_2}{\sigma^2} \right)^{-1} \\ \delta_n &= \tau_n^2 \left[ \frac{\delta_0}{\tau_0^2} + \frac{\sum(y_{i,1} - \mu)}{\sigma^2} - \frac{\sum(y_{i,2} - \mu)}{\sigma^2} \right]\end{aligned}$$

#### 9.4.3 Posterior of $\sigma^2$

**Proposition 9.4.3.**

$$\sigma^2 \mid \mu, \delta, y \sim \text{Inv-Gamma} \left( \frac{\nu_n}{2}, \frac{\nu_n \sigma_n^2}{2} \right)$$

$$\nu_n = \nu_0 + n_1 + n_2$$

$$\nu_n \sigma_n^2 = \nu_0 \sigma_0^2 + \sum (y_{i,1} - (\mu + \delta))^2 + \sum (y_{i,2} - (\mu - \delta))^2$$

---

1. Initialize  $\mu^{(0)}, \delta^{(0)}, \sigma^{2(0)}$

2. For  $t = 1, \dots, T$ :

(a) Sample:

$$\mu^{(t)} \sim p(\mu \mid \delta^{(t-1)}, \sigma^{2(t-1)}, y)$$

(b) Sample:

$$\delta^{(t)} \sim p(\delta \mid \mu^{(t)}, \sigma^{2(t-1)}, y)$$

(c) Sample:

$$\sigma^{2(t)} \sim p(\sigma^2 \mid \mu^{(t)}, \delta^{(t)}, y)$$


---

## 9.5 Gibbs Sampling Algorithm

## 9.6 Exchangeability

**Definition 9.6.1.** A sequence  $Y_1, \dots, Y_n$  is exchangeable if:

$$p(y_1, \dots, y_n) = p(y_{\pi(1)}, \dots, y_{\pi(n)})$$

for any permutation  $\pi$ .

**Remark 9.6.1.** Exchangeability justifies modeling observations as conditionally IID.

## 9.7 Hierarchical Normal Model

### 9.7.1 Model Structure

**Definition 9.7.1.** Within-group model:

$$Y_{ij} \mid \theta_j, \sigma^2 \sim \mathcal{N}(\theta_j, \sigma^2)$$

Between-group model:

$$\theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2)$$

**Remark 9.7.1.** •  $\theta_j$  = group-specific mean

- $\mu$  = global mean
- $\tau^2$  = between-group variance

## 9.8 Interpretation

**Remark 9.8.1.** This model induces **partial pooling**:

- Estimates are shrunk toward the global mean
- Amount of shrinkage depends on  $\tau^2$

## 9.9 Shrinkage Effect

**Proposition 9.9.1.** *Posterior mean of  $\theta_j$  is:*

$$\mathbb{E}[\theta_j \mid y] = w_j \bar{y}_j + (1 - w_j)\mu$$

**Remark 9.9.1.** • If  $\tau^2$  is small  $\rightarrow$  strong pooling

- If  $\tau^2$  is large  $\rightarrow$  weak pooling

## 9.10 Why Hierarchical Models Matter

- Avoid overfitting small groups
- Borrow strength across groups
- Naturally quantify uncertainty
- Handle missing data via latent variables

## 9.11 Summary

- Group comparison extends normal models to structured data
- Parameters become layered (data  $\rightarrow$  group  $\rightarrow$  population)
- Conjugacy enables efficient Gibbs sampling
- Hierarchical models introduce shrinkage and improve estimation stability

# Chapter 10

## Bayesian Linear Models

### 10.1 Model Setup

**Definition 10.1.1.** The Bayesian linear model is:

$$y = X\beta + \epsilon$$
$$\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$$

**Remark 10.1.1.** •  $y \in \mathbb{R}^n$

- $X \in \mathbb{R}^{n \times p}$
- $\beta \in \mathbb{R}^p$
- $\sigma^2 = \text{noise variance}$

### 10.2 Likelihood

**Proposition 10.2.1.**

$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta)\right)$$

### 10.3 Conjugate Prior

**Definition 10.3.1.** Normal–Inverse–Gamma prior:

$$\beta \mid \sigma^2 \sim \mathcal{N}(\beta_0, \sigma^2 V_0)$$
$$\sigma^2 \sim \text{Inv-Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right)$$

**Remark 10.3.1.** • Prior scales with  $\sigma^2$

- Enables closed-form posterior

## 10.4 Posterior Distribution

### 10.4.1 Posterior of $\beta$

**Proposition 10.4.1.**

$$\beta \mid \sigma^2, y \sim \mathcal{N}(\beta_n, \sigma^2 V_n)$$

$$V_n = (V_0^{-1} + X^\top X)^{-1}$$

$$\beta_n = V_n(V_0^{-1}\beta_0 + X^\top y)$$

### 10.4.2 Posterior of $\sigma^2$

**Proposition 10.4.2.**

$$\sigma^2 \mid y \sim \text{Inv-Gamma}\left(\frac{\nu_n}{2}, \frac{\nu_n \sigma_n^2}{2}\right)$$

$$\nu_n = \nu_0 + n$$

$$\nu_n \sigma_n^2 = \nu_0 \sigma_0^2 + (y - X\beta_n)^\top (y - X\beta_n) + (\beta_n - \beta_0)^\top V_0^{-1} (\beta_n - \beta_0)$$

## 10.5 Marginal Posterior of $\beta$

**Proposition 10.5.1.**

$$\beta \mid y \sim t_{\nu_n}(\beta_n, \sigma_n^2 V_n)$$

**Remark 10.5.1.** • arises from integrating out  $\sigma^2$

- heavier tails than normal

## 10.6 Posterior Predictive Distribution

**Proposition 10.6.1.** For a new observation  $y^*$  with covariate  $x^*$ :

$$y^* \mid y \sim t_{\nu_n}\left(x^{*\top} \beta_n, \sigma_n^2 (1 + x^{*\top} V_n x^*)\right)$$

## 10.7 Interpretation

**Remark 10.7.1.** Posterior mean:

$$\beta_n = (V_0^{-1} + X^\top X)^{-1}(V_0^{-1}\beta_0 + X^\top y)$$

**Remark 10.7.2.** This is a weighted average of:

- Prior information ( $\beta_0$ )
- Data information ( $X^\top y$ )

## 10.8 Connection to OLS

**Proposition 10.8.1.** *If prior is diffuse:*

$$V_0^{-1} \rightarrow 0$$

*then:*

$$\beta_n \rightarrow (X^\top X)^{-1} X^\top y$$

**Remark 10.8.1.** Bayesian estimator reduces to OLS.

## 10.9 Shrinkage Interpretation

**Remark 10.9.1.** Bayesian regression performs shrinkage:

- pulls estimates toward  $\beta_0$
- strength depends on  $V_0$

## 10.10 Special Case: Ridge Regression

**Proposition 10.10.1.** *If:*

$$\beta \sim \mathcal{N}(0, \lambda^{-1} I)$$

*then:*

$$\beta_n = (X^\top X + \lambda I)^{-1} X^\top y$$

**Remark 10.10.1.** Bayesian interpretation of ridge regression.

## 10.11 Gibbs Sampling

---

1. Initialize  $\beta^{(0)}, \sigma^{2(0)}$

2. For  $t = 1, \dots, T$ :

(a) Sample:

$$\beta^{(t)} \sim p(\beta \mid \sigma^{2(t-1)}, y)$$

(b) Sample:

$$\sigma^{2(t)} \sim p(\sigma^2 \mid \beta^{(t)}, y)$$


---

## 10.12 Summary

- Linear model extends normal model to multivariate predictors
- Conjugate priors give closed-form posterior
- Posterior is Normal–Inverse–Gamma
- Marginal posterior is Student-t
- Includes OLS and ridge as special cases

# Chapter 11

## Generalized Linear Models (GLMs)

### 11.1 Motivation

**Remark 11.1.1.** Linear models assume:

- Gaussian noise
- Constant variance
- Unbounded response

These assumptions fail for:

- Binary data
- Count data
- Skewed / non-Gaussian responses

### 11.2 GLM Structure

**Definition 11.2.1.** A generalized linear model consists of:

1. Random component:

$$y_i \sim \text{Exponential Family}$$

2. Systematic component:

$$\eta_i = x_i^\top \beta$$

3. Link function:

$$g(\mu_i) = \eta_i$$

**Remark 11.2.1.** •  $\mu_i = \mathbb{E}[y_i \mid x_i]$

- $g(\cdot)$  connects mean to linear predictor



## 11.3 Exponential Family Form

**Definition 11.3.1.**

$$p(y \mid \theta) = h(y) \exp\{\phi(\theta)t(y) - A(\theta)\}$$

**Remark 11.3.1.** GLMs arise because many common distributions (Bernoulli, Poisson, Normal) belong to this family

## 11.4 Canonical Link

**Definition 11.4.1.** A canonical link satisfies:

$$\eta = \phi(\theta)$$

**Remark 11.4.1.** This simplifies likelihood and inference.

## 11.5 Common GLMs

### 11.5.1 Logistic Regression (Binary Data)

**Definition 11.5.1.**

$$y_i \sim \text{Bernoulli}(p_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^\top \beta$$

**Remark 11.5.1.** • Models probabilities in  $(0, 1)$

- Uses logit link

**Proposition 11.5.1.** *Likelihood:*

$$p(y \mid \beta) = \prod_i p_i^{y_i} (1-p_i)^{1-y_i}$$

**Remark 11.5.2.** Appears in your notes as:

$$\text{logit}(p_i) = \alpha + \beta x_i$$

### 11.5.2 Poisson Regression (Count Data)

**Definition 11.5.2.**

$$y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = x_i^\top \beta$$

**Remark 11.5.3.** • Ensures  $\lambda_i > 0$

- Natural for count data

### 11.5.3 Gaussian GLM

**Definition 11.5.3.**

$$y_i \sim \mathcal{N}(x_i^\top \beta, \sigma^2)$$

**Remark 11.5.4.** This reduces to the standard linear model.

## 11.6 Bayesian GLMs

### 11.6.1 Prior

**Definition 11.6.1.**

$$\beta \sim \mathcal{N}(\beta_0, \Sigma_0)$$

### 11.7 Posterior

**Proposition 11.7.1.**

$$p(\beta \mid y) \propto p(y \mid \beta)p(\beta)$$

**Remark 11.7.1.** • No closed form (non-conjugate)

- Requires numerical methods

## 11.8 Inference Methods

### 11.8.1 Laplace Approximation

**Definition 11.8.1.** Approximate posterior by Gaussian around mode:

$$p(\beta \mid y) \approx \mathcal{N}(\hat{\beta}, H^{-1})$$

**Remark 11.8.1.** •  $\hat{\beta}$  = MAP estimate

- $H$  = Hessian of log posterior

### 11.8.2 Metropolis-Hastings

---

1. Propose  $\beta^* \sim q(\cdot \mid \beta^{(t)})$

2. Accept with probability:

$$\min \left( 1, \frac{p(y \mid \beta^*)p(\beta^*)}{p(y \mid \beta^{(t)})p(\beta^{(t)})} \right)$$


---

**Remark 11.8.2.** Used when posterior cannot be sampled directly

### 11.8.3 Gibbs Sampling (Augmentation)

- Remark 11.8.3.**
- Introduce latent variables
  - Convert model into conditionally conjugate form

## 11.9 Interpretation

- Remark 11.9.1.**
- Coefficients act on transformed mean
  - Interpretation depends on link function

### 11.9.1 Logistic Regression

**Proposition 11.9.1.**

$$\beta_j = \text{log-odds change per unit increase in } x_j$$

### 11.9.2 Poisson Regression

**Proposition 11.9.2.**

$$e^{\beta_j} = \text{multiplicative effect on expected count}$$

## 11.10 Summary

- GLMs extend linear models to non-Gaussian data
- Built from exponential family distributions
- Use link functions to ensure valid outputs
- Bayesian GLMs are typically non-conjugate
- Require approximation (Laplace, MCMC, VI)

# Chapter 12

## Bayesian Probit Regression Analysis

### 12.1 Model Setup

**Definition 12.1.1.** Binary response model:

$$y_i \in \{0, 1\}$$

$$\mathbb{P}(y_i = 1 \mid x_i) = \Phi(x_i^\top \beta)$$

**Remark 12.1.1.** •  $\Phi(\cdot)$  = standard normal CDF

- ensures probabilities lie in  $(0, 1)$

### 12.2 Latent Variable Representation

**Definition 12.2.1.** Introduce latent variable:

$$z_i = x_i^\top \beta + \epsilon_i$$

$$\epsilon_i \sim \mathcal{N}(0, 1)$$

$$y_i = \begin{cases} 1 & \text{if } z_i > 0 \\ 0 & \text{if } z_i \leq 0 \end{cases}$$

**Proposition 12.2.1.**

$$\mathbb{P}(y_i = 1 \mid x_i) = \mathbb{P}(z_i > 0) = \Phi(x_i^\top \beta)$$

**Remark 12.2.1.** Probit regression is a thresholded Gaussian linear model.

## 12.3 Likelihood

**Proposition 12.3.1.**

$$p(y \mid \beta) = \prod_{i=1}^n \Phi(x_i^\top \beta)^{y_i} [1 - \Phi(x_i^\top \beta)]^{1-y_i}$$

**Remark 12.3.1.** No conjugate prior exists in this form.

## 12.4 Prior

**Definition 12.4.1.**

$$\beta \sim \mathcal{N}(\beta_0, \Sigma_0)$$

## 12.5 Posterior

**Proposition 12.5.1.**

$$p(\beta \mid y) \propto p(y \mid \beta)p(\beta)$$

**Remark 12.5.1.** Posterior has no closed form  $\rightarrow$  requires sampling.

## 12.6 Data Augmentation

**Definition 12.6.1.** Introduce latent variables:

$$z_i \mid \beta \sim \mathcal{N}(x_i^\top \beta, 1)$$

**Proposition 12.6.1.** *Conditional distributions:*

$$z_i \mid y_i, \beta \sim \begin{cases} \mathcal{N}(x_i^\top \beta, 1) \text{ truncated at } (0, \infty) & y_i = 1 \\ \mathcal{N}(x_i^\top \beta, 1) \text{ truncated at } (-\infty, 0] & y_i = 0 \end{cases}$$

**Remark 12.6.1.** This converts the model into a Gaussian regression problem.

## 12.7 Conditional Posterior of $\beta$

**Proposition 12.7.1.**

$$\beta \mid z \sim \mathcal{N}(\beta_n, V_n)$$

$$V_n = (X^\top X + \Sigma_0^{-1})^{-1}$$

$$\beta_n = V_n(X^\top z + \Sigma_0^{-1}\beta_0)$$

**Remark 12.7.1.** Conditional on  $z$ , this is identical to Bayesian linear regression.

## 12.8 Gibbs Sampling Algorithm

- 
1. Initialize  $\beta^{(0)}$
  2. For  $t = 1, \dots, T$ :
    - (a) Sample latent variables:
 
$$z_i^{(t)} \sim p(z_i \mid y_i, \beta^{(t-1)})$$
    - (b) Sample coefficients:
 
$$\beta^{(t)} \sim p(\beta \mid z^{(t)})$$
- 

**Remark 12.8.1.** • Fully Gibbs  $\rightarrow$  no MH needed

- Efficient and stable

## 12.9 Posterior Prediction

**Proposition 12.9.1.**

$$\mathbb{P}(y^* = 1 \mid x^*, \beta) = \Phi(x^{*\top} \beta)$$

**Remark 12.9.1.** Posterior predictive:

$$\mathbb{P}(y^* = 1 \mid y) = \int \Phi(x^{*\top} \beta) p(\beta \mid y) d\beta$$

## 12.10 Interpretation

**Remark 12.10.1.** •  $\beta_j$  affects latent variable  $z$

- not directly interpretable as odds (unlike logistic)

## 12.11 Comparison with Logistic Regression

**Proposition 12.11.1.** • *Logistic: logit link*

- *Probit: normal CDF link*

**Remark 12.11.1.** • Probit allows exact Gibbs via augmentation

- Logistic typically requires Metropolis or specialized augmentation

## 12.12 Summary

- Probit model uses Gaussian latent variable
- Binary outcomes arise via thresholding
- Data augmentation restores conjugacy
- Enables efficient Gibbs sampling
- Closely related to linear Gaussian models

# Chapter 13

## Missing Data and Imputation

### 13.1 Motivation

**Remark 13.1.1.** Real datasets are rarely complete.

Naive approaches:

- Delete missing observations (information loss)
- Fill with mean or constants (bias)

Bayesian approach:

- Treat missing values as *unknown random variables*
- Integrate or sample them as part of inference

### 13.2 Data Decomposition

**Definition 13.2.1.** Partition data:

$$Y = (Y_{\text{obs}}, Y_{\text{miss}})$$

**Proposition 13.2.1.** *Target posterior:*

$$p(\theta \mid Y_{\text{obs}}) = \int p(\theta, Y_{\text{miss}} \mid Y_{\text{obs}}) dY_{\text{miss}}$$

**Remark 13.2.1.** Integration is usually intractable  $\rightarrow$  use sampling.

### 13.3 Missing Data Mechanisms

**Definition 13.3.1.** MCAR (Missing Completely at Random):

$$p(R \mid Y_{\text{obs}}, Y_{\text{miss}}, \theta) = p(R)$$

MAR (Missing at Random):

$$p(R \mid Y_{\text{obs}}, Y_{\text{miss}}, \theta) = p(R \mid Y_{\text{obs}})$$

MNAR (Not Missing at Random): depends on  $Y_{\text{miss}}$



**Remark 13.3.1.** • MCAR and MAR  $\rightarrow$  standard Bayesian imputation valid  
 • MNAR  $\rightarrow$  must explicitly model missingness

## 13.4 Bayesian Formulation

**Proposition 13.4.1.** *Joint posterior:*

$$p(\theta, Y_{\text{miss}} \mid Y_{\text{obs}}) \propto p(Y_{\text{obs}}, Y_{\text{miss}} \mid \theta) p(\theta)$$

**Remark 13.4.1.** Missing values are treated exactly like latent variables.

## 13.5 Data Augmentation

**Remark 13.5.1.** Same idea as probit:

*Unknown quantities  $\rightarrow$  introduce  $\rightarrow$  sample  $\rightarrow$  iterate*

**Definition 13.5.1.** Augmented sampling scheme:

- Treat  $Y_{\text{miss}}$  as parameters
- Sample them along with  $\theta$

## 13.6 Gibbs Sampling with Missing Data

---

At iteration  $t$ :

1. Sample parameters:

$$\theta^{(t)} \sim p(\theta \mid Y_{\text{obs}}, Y_{\text{miss}}^{(t-1)})$$

2. Sample missing data:

$$Y_{\text{miss}}^{(t)} \sim p(Y_{\text{miss}} \mid Y_{\text{obs}}, \theta^{(t)})$$


---

**Remark 13.6.1.** This is a direct extension of:

- Probit latent variable sampling
- General Gibbs sampling

## 13.7 Key Insight

**Proposition 13.7.1.** *Imputation is sampling from:*

$$p(Y_{\text{miss}} \mid Y_{\text{obs}}, \theta)$$

**Remark 13.7.1.** • Not a single “best guess”

- A full distribution of plausible values

## 13.8 Example: Normal Model with Missing Data

**Definition 13.8.1.** Assume:

$$y_i \sim \mathcal{N}(\mu, \sigma^2)$$

**Proposition 13.8.1.** *Conditional imputation:*

$$y_i^{\text{miss}} \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2)$$

**Remark 13.8.1.** Missing values are sampled exactly like observed ones.

## 13.9 Multiple Imputation

**Definition 13.9.1.** Instead of one dataset:

- Generate multiple completed datasets
- Analyze each
- combine results

**Remark 13.9.1.** Captures uncertainty from missingness.

## 13.10 Complete Case vs Bayesian Approach

**Proposition 13.10.1.** • *Complete case:*

- *biased unless MCAR*
- *wastes data*

- *Bayesian:*
  - *uses all available information*
  - *propagates uncertainty*

## 13.11 Connection to Earlier Topics

**Remark 13.11.1.** • Probit: latent  $z_i$

- Missing data: latent  $Y_{\text{miss}}$
- Both use:
  - augmentation
  - Gibbs sampling

## 13.12 General Template

1. Initialize missing values
2. For  $t = 1, \dots, T$ :
  - (a) Sample  $\theta^{(t)}$
  - (b) Sample  $Y_{\text{miss}}^{(t)}$
3. Use samples for inference

## 13.13 Summary

- Missing data treated as latent variables
- Posterior integrates over missing uncertainty
- Gibbs sampling enables tractable inference
- Avoids bias from ad-hoc imputation
- Unifies with latent variable modeling

# Chapter 14

## GLM and Beyond

### 14.1 Limitations of GLMs

**Remark 14.1.1.** Generalized Linear Models assume:

- Correct link function
- Correct distributional form
- Independent observations

These assumptions may fail in practice.

### 14.2 Non-Conjugacy

**Remark 14.2.1.** For most GLMs:

- Posterior distributions are not available in closed form
- Conjugate priors do not exist

**Proposition 14.2.1.**

$$p(\beta \mid y) \propto p(y \mid \beta)p(\beta)$$

**Remark 14.2.2.** Requires approximation or simulation methods.

### 14.3 Laplace Approximation

**Definition 14.3.1.** Approximate posterior around mode:

$$p(\beta \mid y) \approx \mathcal{N}(\hat{\beta}, H^{-1})$$

**Remark 14.3.1.**

- $\hat{\beta}$  = MAP estimate
- $H$  = Hessian of log posterior

## 14.4 Data Augmentation

**Remark 14.4.1.** Introduce latent variables to simplify models:

- Probit: latent Gaussian variable
- Missing data: latent observations

**Remark 14.4.2.** Transforms non-conjugate models into conditionally conjugate ones.

## 14.5 Hierarchical GLMs

**Definition 14.5.1.** Extend GLMs with group structure:

$$y_{ij} \mid \theta_j \sim \text{GLM}$$

$$\theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2)$$

**Remark 14.5.1.** Allows:

- Partial pooling
- Modeling heterogeneity

## 14.6 Overdispersion

**Remark 14.6.1.** Poisson assumption:

$$\text{Var}(Y) = \mathbb{E}[Y]$$

Often violated in practice.

**Proposition 14.6.1.** *Alternative:*

- *Negative Binomial model*
- *Random effects models*

## 14.7 Robust Models

**Remark 14.7.1.** Normal errors are sensitive to outliers.

Alternatives:

- Student-t likelihood
- Heavy-tailed priors

## 14.8 Nonparametric Extensions

**Remark 14.8.1.** Relax parametric assumptions:

- Dirichlet Process models
- Mixture models

## 14.9 Summary

- GLMs extend linear models to non-Gaussian data
- Non-conjugacy requires approximation methods
- Data augmentation restores tractability
- Hierarchical extensions capture complex structure
- Robust and nonparametric models improve flexibility

# Chapter 15

## Model Selection

### 15.1 Motivation

**Remark 15.1.1.** Given competing models  $M_1, M_2, \dots$ , we want to determine which model best explains the data.

### 15.2 Bayesian Model Comparison

**Definition 15.2.1.** Posterior model probability:

$$p(M_i | y) = \frac{p(y | M_i)p(M_i)}{\sum_j p(y | M_j)p(M_j)}$$

### 15.3 Marginal Likelihood

**Definition 15.3.1.**

$$p(y | M) = \int p(y | \theta, M)p(\theta | M) d\theta$$

**Remark 15.3.1.** Accounts for:

- Model fit
- Model complexity

### 15.4 Bayes Factors

**Definition 15.4.1.** For models  $M_1$  and  $M_2$ :

$$BF_{12} = \frac{p(y | M_1)}{p(y | M_2)}$$

**Remark 15.4.1.** •  $BF > 1$  favors  $M_1$

- $BF < 1$  favors  $M_2$

## 15.5 Interpretation

**Remark 15.5.1.** •  $1 \leq BF < 3$ : weak evidence

- $3 \leq BF < 10$ : moderate evidence
- $10 \leq BF < 100$ : strong evidence
- $BF \geq 100$ : decisive

## 15.6 Important Considerations

**Remark 15.6.1.** • Requires proper priors

- Sensitive to prior choice
- Penalizes model complexity automatically

## 15.7 Marginalization Paradox

**Remark 15.7.1.** If prior is improper:

$$p(y \mid M) \text{ is undefined}$$

Therefore Bayes factors cannot be computed.

## 15.8 Alternative Criteria

### 15.8.1 BIC

**Definition 15.8.1.**

$$\text{BIC} = -2 \log p(y \mid \hat{\theta}) + k \log n$$

**Remark 15.8.1.** • Penalizes complexity

- Large sample approximation to Bayes factor

### 15.8.2 Deviance Information Criterion (DIC)

**Definition 15.8.2.**

$$\text{DIC} = \bar{D} + p_D$$

**Remark 15.8.2.** •  $\bar{D}$  = posterior mean deviance

- $p_D$  = effective number of parameters

### 15.8.3 Log Pseudo-Marginal Likelihood (LPML)

**Remark 15.8.3.** Based on leave-one-out predictive densities.



## 15.9 Posterior Predictive Checks

**Definition 15.9.1.** Compare observed data to simulated data:

$$y^{rep} \sim p(y^{rep} \mid y)$$

**Remark 15.9.1.** Detects model misfit.

## 15.10 Tradeoff: Fit vs Complexity

**Remark 15.10.1.** Good models:

- Fit data well
- Avoid overfitting

## 15.11 Summary

- Model selection compares competing explanations of data
- Bayes factors use marginal likelihoods
- Proper priors are required
- Alternative criteria approximate model evidence
- Balance between fit and complexity is essential